📄 AQI Predictor using Machine Learning – Detailed Project Report

---

1. Introduction

Air pollution has become one of the most serious environmental and public health challenges in recent years. Rapid industrialization, urbanization, and increasing vehicular emissions have significantly contributed to the degradation of air quality. The *Air Quality Index (AQI)* is a standardized indicator used to measure the level of air pollution and its potential impact on human health. It converts complex air pollutant data into a single numerical value that is easy to understand. Monitoring AQI helps individuals and authorities take preventive measures to reduce exposure to harmful pollutants. This project focuses on using machine learning techniques to predict AQI values based on pollutant concentrations. By analysing historical pollutant data, the system can estimate future AQI levels, thereby helping in awareness and decision-making. The project demonstrates how data-driven approaches can be applied to solve real-world environmental problems efficiently.

---

2. Problem Statement

The main objective of this project is to design and implement a system that can accurately predict the Air Quality Index using pollutant data. In real-world scenarios, AQI is influenced by multiple pollutants such as PM2.5, PM10, NO_2, SO_2, NH_3, CO, and Ozone. Understanding the relationship between these pollutants and AQI can help in forecasting pollution levels. The problem lies in developing a predictive model that can learn from existing data and provide reliable AQI estimates for new inputs. Traditional methods of AQI calculation can be complex and require domain-specific knowledge. Therefore, this project aims to simplify the process using machine learning. The system takes pollutant values as input and produces a numerical AQI output. This can be useful for individuals,

researchers, and policymakers to assess air quality conditions quickly and take necessary actions.

---

3. Dataset

The dataset used in this project contains pollutant concentration values along with corresponding AQI values. The key features include PM2.5, PM10, NO_2, SO_2, NH_3, CO, and Ozone, which are major contributors to air pollution. Each row in the dataset represents a specific observation where pollutant levels and AQI are recorded. The dataset plays a crucial role in training the machine learning model, as it helps the algorithm understand patterns and relationships between input features and the target variable. Before training, the dataset is checked for missing or inconsistent values and cleaned accordingly to ensure accuracy. Proper preprocessing is important to improve model performance. The dataset used in this project is structured and numerical, making it suitable for regression-based machine learning algorithms. Overall, the dataset serves as the foundation for building an effective AQI prediction system.

---

4. Methodology

The methodology of this project involves several steps starting from data preprocessing to model training and evaluation. First, the dataset is loaded and inspected to understand its structure. Any missing values are handled using appropriate techniques such as removal or imputation. Next, relevant features are selected, and the data is split into training and testing sets to evaluate model performance. A *Random Forest Regression* algorithm is used for prediction, as it is robust and provides good accuracy for numerical data. The model is trained using the training dataset and then tested on unseen data to measure its performance. Evaluation metrics such as Mean Absolute Error (MAE) and R^2 Score are used to assess accuracy. The trained model is then saved for future use. This systematic approach ensures that the model is reliable and performs well on new inputs.

5. Implementation

The implementation of this project is carried out using Python and various machine learning libraries. Pandas is used for data handling and preprocessing, while NumPy is used for numerical operations. The Scikit-learn library is used to implement the Random Forest Regression model. The dataset is loaded into a DataFrame, and features and target variables are defined. The model is trained using the training data and evaluated using test data. Once trained, the model is saved as a .pkl file using the pickle module. A separate Python script is created to take user input from the terminal. The user enters pollutant values, and the model predicts the corresponding AQI value. This terminal-based interface ensures simplicity and ease of use. The implementation demonstrates how machine learning models can be integrated into simple applications for real-world usage.

6. Results

The results of this project show that the machine learning model is capable of predicting AQI values based on pollutant inputs. After training the model, predictions are made on the test dataset, and the performance is evaluated using metrics such as Mean Absolute Error and R^2 Score. The results indicate that the model can reasonably estimate AQI values, demonstrating a strong relationship between pollutant concentrations and air quality. When users input pollutant values through the terminal, the system successfully generates AQI predictions in real-time. Although the predictions may not be perfectly accurate due to limitations in data and external factors, they provide a close approximation of actual AQI values. Overall, the results validate the effectiveness of using machine learning techniques for environmental data analysis and prediction.

7. Conclusion

In conclusion, this project successfully demonstrates the application of machine learning in predicting air quality levels. By using pollutant data as input, the system can estimate AQI values, which can help in understanding pollution conditions. The use of the Random Forest Regression model ensures good accuracy and reliability. This project highlights the importance of data-driven approaches in solving real-world problems. It also provides a foundation for further enhancements, such as integrating real-time data and improving prediction accuracy. The system developed is simple, efficient, and easy to use, making it suitable for academic and practical purposes. Overall, the project achieves its objective of building an AQI prediction system using machine learning techniques.

---

8. Limitations

Despite its effectiveness, the project has certain limitations. The accuracy of the model heavily depends on the quality and size of the dataset used for training. Since AQI is influenced by multiple environmental and geographical factors, the model may not capture all real-world variations. The dataset used may not represent all possible scenarios, leading to approximation errors. Additionally, the model does not consider dynamic factors such as weather conditions, time of day, and seasonal variations, which can significantly impact air quality. The system also relies on user input, which may not always be accurate. These limitations highlight the need for more comprehensive datasets and advanced models for improved performance.

---

9. Future Scope

The project can be further improved in several ways. One major enhancement is the integration of real-time data using APIs, which would allow the system to provide live AQI predictions. The model can also be

improved by using advanced algorithms such as XGBoost or Deep Learning techniques. Another improvement is the development of a graphical user interface (GUI) or web application to make the system more user-friendly. Additionally, incorporating weather data such as temperature, humidity, and wind speed can improve prediction accuracy. Expanding the dataset to include data from different locations and time periods can also enhance model performance. These improvements can make the system more robust and practical for real-world applications.